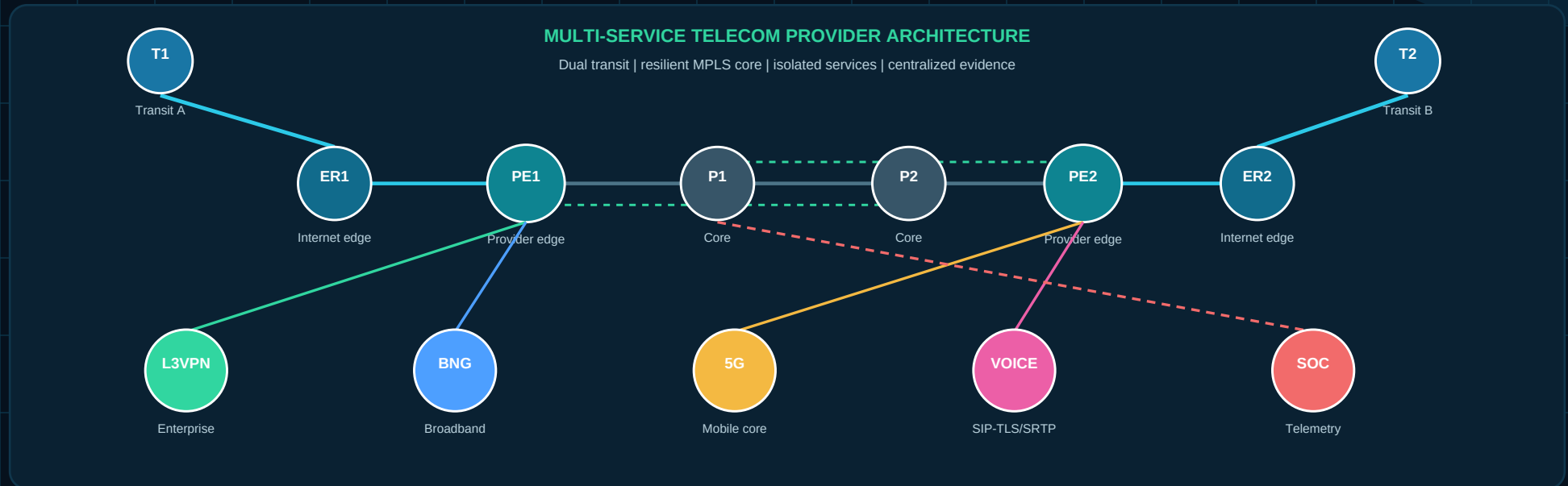


Telecom + MPLS + 5G Cybersecurity Super Lab

A multi-service provider architecture combining resilient routing, service isolation, telecom workloads, security controls, observability, and evidence-driven validation



5 SERVICE VRFS	35 FAULT / SECURITY SCENARIOS	30 DETECTION USE CASES	388 STATIC ASSERTIONS
Enterprise Broadband Mobile Voice Internet	Failure injection and recovery evidence	Splunk Sentinel LogScale	0 errors 0 warnings

Prepared by Marc Lavoie | Technical portfolio report | August 2026

Executive engineering brief

This lab models a service-provider environment in which routing, MPLS transport, subscriber access, mobile core, voice, Internet edge, and SOC workflows are designed as one operational system. The package is intentionally evidence-led: every service has prerequisites, acceptance gates, failure scenarios, and expected proof.

185	5	35	35	30	388
FILES	SERVICE VRFS	SESSIONS / SERVICES	FAILURE SCENARIOS	DETECTION USE CASES	STATIC ASSERTIONS

ARCHITECTURE Dual transit edges, resilient provider core, route reflectors, redundant provider edges, enterprise VPN, broadband, 5G SA, secure voice, and SOC/NOC management.	SECURITY Control-plane protection, VRF boundaries, subscriber trust, telecom encryption, management hardening, telemetry collection, and platform-specific detection content.	VALIDATION Structured topology/IPAM checks, routing assertions, protocol checks, service-path gates, failure recovery scenarios, and an explicit evidence standard.
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Why this project matters

- Connects carrier routing and telecom services to practical defensive-security outcomes.
- Separates provider transport, customer routes, subscriber services, management, and telemetry through clear trust boundaries.
- Uses repeatable validation instead of relying on screenshots or configuration volume as proof of correctness.
- Documents degraded states and recovery criteria so the lab can be operated, not merely assembled.

INDEPENDENT STATIC GATE
The supplied package hash matched the documented release. The complete validation runner reported 185 files, 388 configuration and topology assertions, 0 errors, and 0 warnings.

SCOPE BOUNDARY
This result proves package structure and cross-file consistency. Live device boot, protocol convergence, traffic generation, packet capture, and recovery timing remain runtime evidence tasks.

Integrated service-provider architecture

The full topology ties the provider control plane to customer and telecom services. Solid links represent service and routing relationships; dashed paths represent telemetry or management flows.

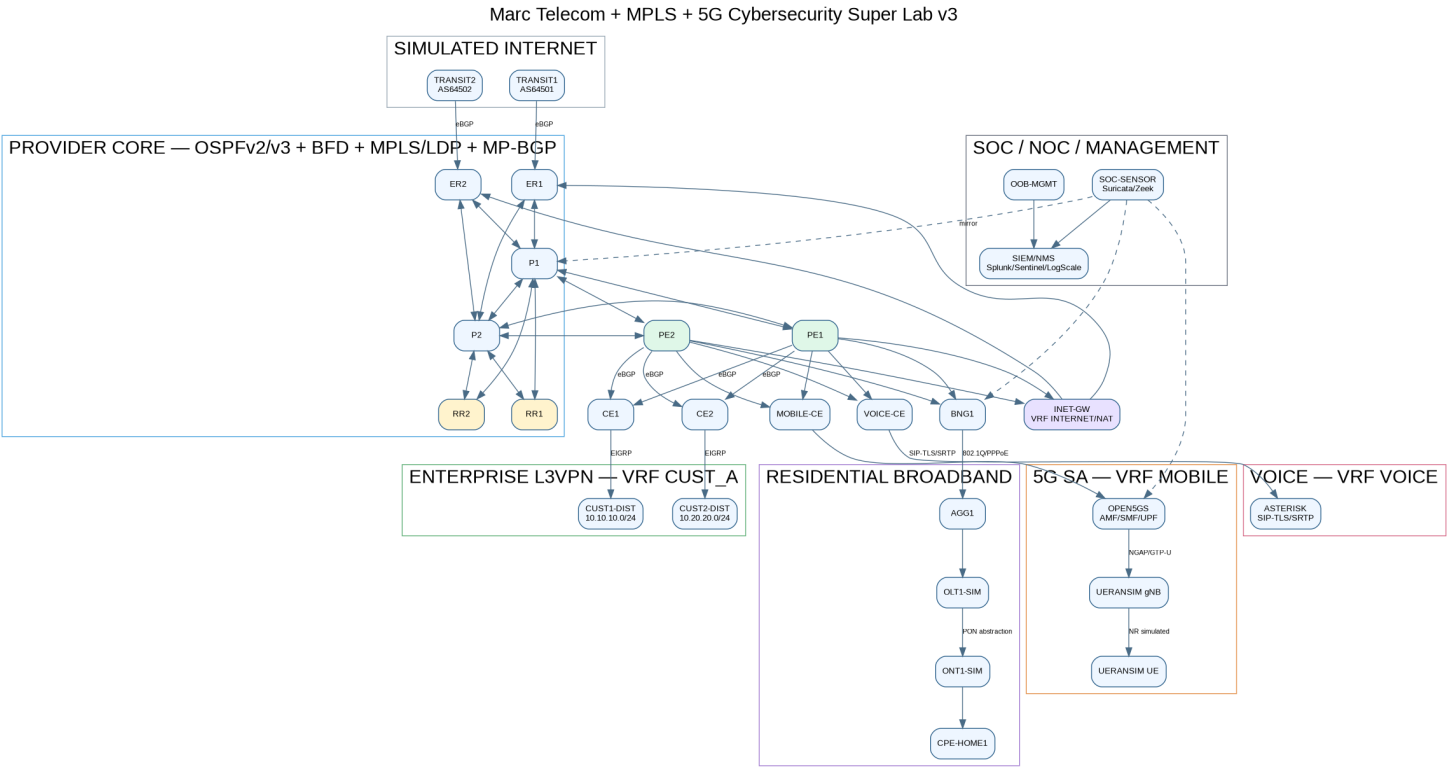


Figure 1. Source architecture diagram from the validated lab package. All addressing and subscriber data remain within the private lab artifact.

Domain	Primary components	Engineering purpose
Provider core	ER1/ER2, P1/P2, PE1/PE2, RR1/RR2	Resilient underlay, label switching, Internet edge, and VPN route distribution.
Enterprise	CE routers and two customer sites	Isolated Layer-3 VPN service with independent customer routing.
Broadband	BNG, aggregation, access, subscriber CPE	PPPoE access, subscriber policy, authentication, and Internet service.
Mobile	Open5GS and UERANSIM	5G standalone control and user-plane study environment.
Voice	Asterisk	Protected signaling and media workflow using SIP-TLS and SRTP.
SOC / NOC	OOB management, sensors, SIEM	Centralized control, telemetry, investigation, and evidence collection.

Provider core and control-plane resilience

The diagram illustrates a Multi-Service Telecom Provider Architecture. At the top, the title "MULTI-SERVICE TELECOM PROVIDER ARCHITECTURE" is displayed in blue, followed by the tagline "Dual transit | resilient MPLS core | isolated services | centralized evidence". The architecture is composed of several key components and their interconnections:

- Transit Providers:** T1 (Transit A) and T2 (Transit B) are represented by dark blue circles at the top corners.
- Internet Edges:** ER1 (Internet edge) and ER2 (Internet edge) are dark blue circles connected to T1 and T2 respectively by solid cyan lines.
- Provider Edges:** PE1 (Provider edge) and PE2 (Provider edge) are teal circles connected to ER1 and ER2 respectively by solid cyan lines.
- Core:** P1 (Core) and P2 (Core) are dark blue circles connected to PE1 and PE2 respectively by solid grey lines. P1 and P2 are also connected to each other by a solid grey line.
- Services:** L3VPN (Enterprise, green circle) is connected to PE1 by a solid green line. BNG (Broadband, blue circle) is connected to PE1 by a solid blue line. 5G (Mobile core, orange circle) is connected to PE2 by a solid orange line. VOICE (SIP-TLS/SRTP, pink circle) is connected to PE2 by a solid pink line. SOC (Telemetry, red circle) is connected to PE2 by a solid red line.
- Interconnections:** Dashed lines represent specific interconnections: green dashed lines connect PE1 to P1 and P2; red dashed lines connect P1 and P2 to PE2; and a red dashed line connects P1 to SOC.

Control-plane hardening		
ROUTING AUTHENTICITY Peer allowlists, maximum-prefix policy, route filtering, RPKI study controls, and explicit import/export intent.	MANAGEMENT Out-of-band administration, SSH, SNMPv3, protected management APIs, time synchronization, and logged change activity.	RESILIENCE Redundant P/PE paths, dual transit, route reflectors, convergence checks, and rollback-oriented failure scenarios.

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Five VRFs define the service boundary

Each service receives a distinct forwarding and policy context. Route targets make intended membership explicit and reduce the chance that enterprise, subscriber, mobile, voice, or Internet routes leak across trust zones.

VRF	Service	Primary trust concern	Validation focus
CUST_A	Enterprise L3VPN	Customer route isolation and correct site membership.	Import/export policy and end-to-end customer reachability.
BROADBAND	Residential Internet	Subscriber separation, identity, and policy enforcement.	Session establishment, AAA result, address assignment, and Internet path.
MOBILE	5G standalone	Subscriber trust, control-plane integrity, and user-plane separation.	Registration state, session creation, tunnel path, and policy.
VOICE	Protected voice	Signaling authentication and media confidentiality.	SIP-TLS establishment, SRTP media, and call evidence.
INTERNET	Shared Internet edge	Route hygiene, NAT boundary, and external exposure.	Default path, dual-transit behavior, translation, and filtering.

Service-chain logic

INGRESS Identify the subscriber, customer, device, or service edge and place traffic in the correct context.	POLICY Apply routing, authentication, filtering, translation, and encryption controls appropriate to that context.	TRANSPORT Carry the service through the MPLS core without exposing unrelated customer or service routes.	EVIDENCE Collect control-plane state, packet evidence, telemetry, and SIEM artifacts tied to the same test case.
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Isolation acceptance checks

- Only intended route targets are imported into each service VRF.
- Provider core nodes do not require customer or subscriber routes to forward labeled traffic.
- Shared Internet access is explicit and does not become an uncontrolled route-leak shortcut.
- Management and telemetry paths remain distinct from production service forwarding.

DESIGN PRINCIPLE

A successful ping is not sufficient evidence. The route must appear in the correct table, traverse the intended service path, and be absent from unrelated VRFs.

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Broadband, 5G, and protected voice

The telecom layers are represented as explicit service paths with prerequisites, security controls, and evidence gates rather than as isolated application demos.

Service path	Workflow	Security and evidence
Residential broadband	Access -> aggregation -> BNG -> subscriber policy -> Internet edge	PPPoE session, RADIUS decision, assigned service context, NAT path, and subscriber isolation.
5G standalone	UE/RAN simulator -> 5G core control plane -> user-plane function -> data network	Subscriber authentication, protected control plane, tunnel establishment, policy, and user-plane telemetry.
Protected voice	Endpoint -> SIP-TLS signaling -> Asterisk -> SRTP media	Certificate trust, signaling state, encrypted media, call completion, and bounded logging.

BROADBAND Baseline uses a documented access VLAN. QinQ is retained as an optional profile when the chosen images and host platform can prove it.	5G SA Open5GS and UERANSIM provide a contained study environment. Public documentation omits subscriber test secrets and unique identifiers.	VOICE SIP-TLS protects signaling and SRTP protects media. Validation must prove both the call and the protection state.
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Telecom acceptance sequence

Stage	Question	Minimum evidence
Identity	Was the subscriber or endpoint recognized under the intended policy?	Authentication result and policy association.
Session	Did the requested broadband, mobile, or voice session form?	State transition and endpoint/service confirmation.
Transport	Did traffic follow the intended provider and service path?	Route/label/tunnel state plus packet or flow evidence.
Protection	Were the required trust and encryption controls active?	Protocol state, certificate/association evidence, and negative test.
Recovery	Did the service return to a known-good state after a fault?	Timestamped before/after evidence and cleanup verification.

Security architecture and trust planes

The lab maps transport, service, management, identity, encryption, and detection controls to concrete evidence. Controls are selected by risk and platform capability, then labeled honestly when they are represented or study-only.

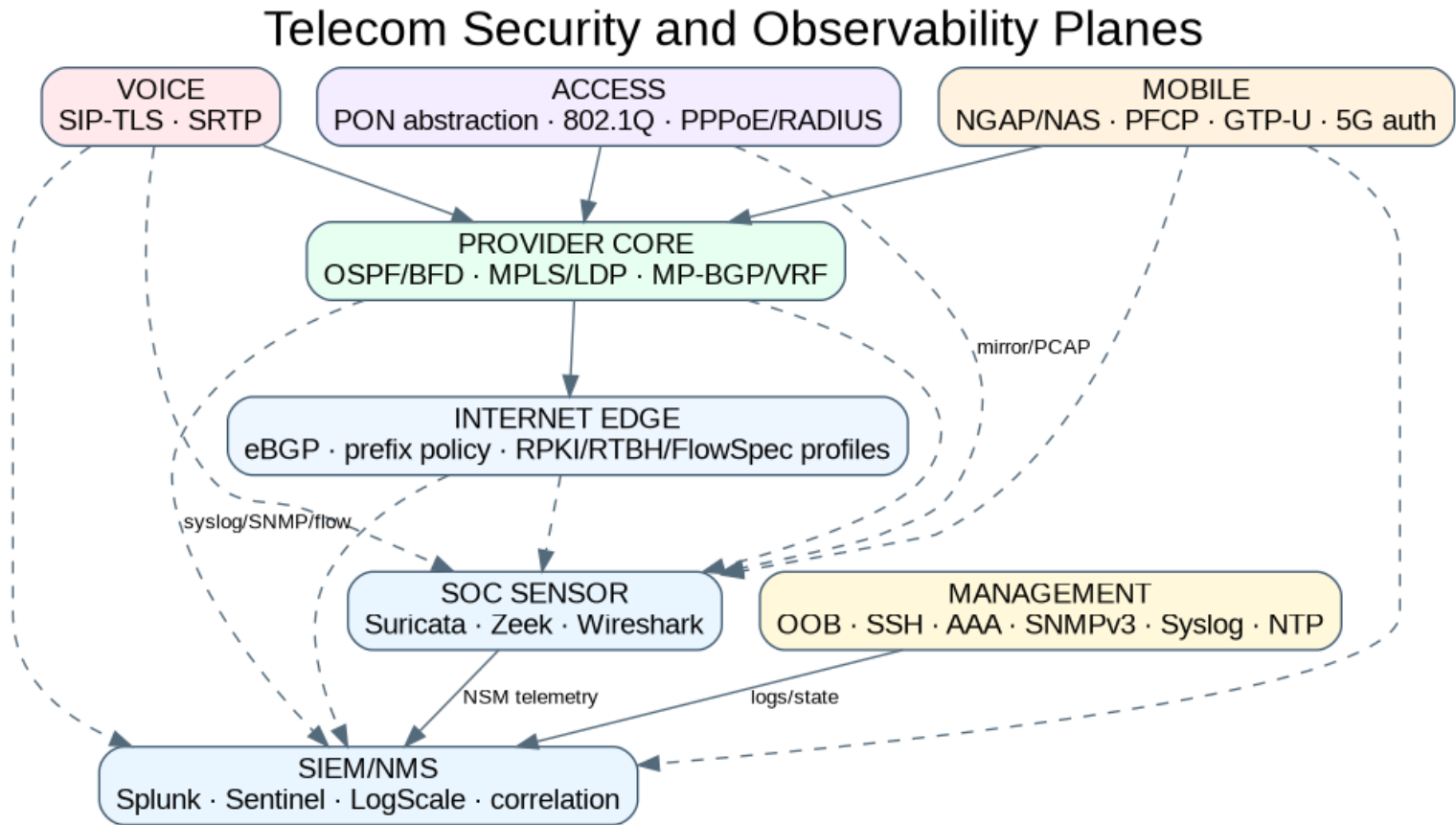


Figure 2. Telecom security-plane diagram supplied with the validated package.

Security plane	Representative controls	Proof expected
Routing and transport	Peer policy, prefix limits, route filtering, LDP scope, VRF boundaries	Neighbor state, route policy result, and isolation tests.
Telecom trust	5G-AKA/SUCI study, protected signaling, encrypted media, subscriber AAA	Authentication, protocol state, packet evidence, and negative tests.
Management	OOB access, SSH, SNMPv3, protected APIs, synchronized time	Access logs, rejected unauthorized paths, and clock consistency.
Detection	Suricata, Zeek, packet capture, SIEM searches, ATT&CK; mapping	Timestamped telemetry, correlated search, finding, and disposition.

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Detection engineering across three SIEM languages

Thirty detection use cases are translated into Splunk SPL, Microsoft Sentinel KQL, and CrowdStrike LogScale templates. The useful artifact is the detection lifecycle: prerequisite telemetry, query logic, validation action, triage context, and expected response.

30	90	3	1
USE CASES	QUERY TEMPLATES	SIEM LANGUAGES	SHARED EVIDENCE MODEL

Detection family	Example signal	Analyst outcome
Provider control plane	Unexpected BGP state, route-policy change, LDP loss, or adjacency churn	Distinguish maintenance, fault, and unauthorized routing behavior.
Subscriber services	Repeated authentication failure, session anomalies, or unusual policy assignment	Correlate subscriber identity, access path, and service outcome.
5G core	Registration failure, control-plane errors, tunnel disruption, or trust mismatch	Identify the failed stage and separate configuration from adversarial activity.
Protected voice	Signaling failures, trust problems, media mismatch, or abnormal call patterns	Validate signaling/media protection and investigate endpoint behavior.
Management plane	Rejected access, configuration change, suspicious command, or time drift	Confirm authorization, scope impact, and evidence integrity.

Evidence-driven detection workflow

1 GENERATE Run a bounded, documented test action with a known start time and expected telemetry source.	2 OBSERVE Confirm raw packet, flow, protocol, or system evidence before evaluating the SIEM query.	3 CORRELATE Execute the platform query and connect the result to the test case, asset role, and service path.	4 DISPOSE Record true/false positive reasoning, response action, rollback, and post-test verification.
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PORTABLE ENGINEERING

The platform syntax changes, but the analytic intent, telemetry prerequisites, validation method, and response decision should remain consistent.

Failure engineering and recovery

Thirty-five scenarios exercise routing, service, trust, telemetry, and management failures. Each scenario is useful only when the expected blast radius, evidence, rollback, and recovered state are defined before execution.

Fault domain	Representative test	Expected operational lesson
Core convergence	Provider link loss, BFD transition, OSPF/LDP disruption	Measure dependency order and verify the alternate path restores service.
VPN control	Route-target mismatch, route-reflector loss, edge-session failure	Identify control-plane state that causes selective service loss or route leakage.
Broadband	AAA rejection, subscriber pool exhaustion, access-path failure	Trace identity, session, address, policy, and forwarding as separate checkpoints.
5G	PLMN mismatch, PFCP/GTP-U disruption, SBI trust failure	Separate registration, control-plane, and user-plane failure domains.
Voice	SIP trust error, SRTP mismatch, service interruption	Prove signaling, media, and application state independently.
Security / operations	Management attack, RPKI issue, timing drift, sensor loss	Preserve evidence quality while containing or restoring the affected plane.

Scenario evidence contract

PRECONDITION Known-good service state, expected neighbors, routes, sessions, clocks, and telemetry.	INJECTION One bounded change with recorded start time, owner, safety boundary, and rollback command.	OBSERVATION Control-plane, data-plane, service, and SIEM evidence tied to the same timeline.	RECOVERY Restore configuration, verify state, compare baselines, and record any residual artifacts.
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Definition of done

- The fault produces the expected and bounded blast radius.
- The monitoring path records enough evidence to explain the failure.
- The documented rollback restores all affected service and security states.
- Post-recovery verification shows no hidden route, session, credential, or telemetry residue.

Validation system and evidence standard

The package validator cross-checks topology, inventory, addressing, routing, services, scenarios, query coverage, and credential hygiene. Runtime validation then adds device output and traffic evidence to the same acceptance model.

Static validation domain	Assertions passed
Interface and IPAM consistency	106
BGP sessions and policy relationships	58
OSPFv2/OSPFv3, BFD, and LDP relationships	130
VRF and shared-Internet design	45
Broadband, AAA, 5G, voice, and DNS	49
Total	388

VALIDATION RESULT

All package YAML parsed successfully. The topology, sessions, service inventory, detection coverage, failure-scenario coverage, and credential-hygiene checks completed with 0 errors and 0 warnings.

Runtime evidence still required

Evidence class	Examples	Acceptance rule
Configuration	Sanitized running configuration and selected control state	Must match the documented role and exclude secrets.
Protocol	Neighbor summaries, route tables, labels, tunnels, registrations, sessions	Must show both success state and relevant policy context.
Traffic	Packet capture, flow record, sourced probe, service transaction	Must prove the intended path and protection state.
Detection	Raw event, SIEM result, analytic explanation, response disposition	Must correlate to the test timeline without exposing private data.
Recovery	Before/fault/after state with timestamps	Must demonstrate complete restoration and residual-artifact checks.

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Truth in emulation

A portfolio lab is strongest when each capability is labeled by what the selected images and environment can actually prove. This project separates design intent from validated behavior so optional features do not become accidental claims.

Status	Meaning	Portfolio treatment
Baseline	Required for the core architecture and expected to run on the chosen platform.	Must have live state and traffic evidence before being described as operational.
Image-dependent baseline	Core intent is required, but exact syntax or feature support varies by network image.	Document the tested image, version, syntax, and observed limitations.
Optional profile	Useful extension that is not required for the primary acceptance gate.	Show separately and never imply it was part of the baseline run.
Represented	The design and evidence contract are documented, but the underlying platform may not implement the function natively.	Label as architecture or workflow representation.
Study only	Security or telecom mechanism is analyzed without claiming a live implementation.	Present threat model, trust boundary, and validation plan only.

Public-safety publishing boundary

EXCLUDED Passwords, tokens, subscriber keys, unique telecom identifiers, private host data, operational logs, and sensitive screenshots.	INCLUDED Architecture roles, protocol responsibilities, security controls, validation results, synthetic diagrams, and evidence requirements.	NEXT STAGE Boot the selected images, capture sanitized live proof, exercise failures, and append measured recovery results.
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PROFESSIONAL STANDARD

The project distinguishes 'designed,' 'statically validated,' and 'runtime proven.' That distinction makes the work more credible to reviewers and safer to publish.

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Recruiter takeaway and next evidence stage

This project demonstrates the ability to reason across network engineering, telecom services, and defensive security while maintaining testability, change control, and honest capability boundaries.

CORE	ACCESS	MOBILE	VOICE	SOC
MPLS / BGP / OSPF	PPPOE / AAA	5G SA	SIP-TLS / SRTP	3 SIEM LANGUAGES

Competency	Evidence in this project
Carrier architecture	Resilient core, route reflection, dual Internet transit, multiple provider edges, and service-specific forwarding contexts.
Network security	Routing policy, trust boundaries, management controls, encryption mapping, and explicit isolation acceptance tests.
Telecom systems	Broadband subscriber workflow, 5G standalone service chain, secure voice, and service-specific failure analysis.
Detection engineering	Thirty analytics expressed in three query languages with shared telemetry and response intent.
Validation automation	A zero-error static gate spanning configuration, topology, addressing, protocols, services, scenarios, and credential hygiene.
Operational discipline	Preconditions, bounded fault injection, rollback, timestamped evidence, and post-recovery verification.

Recommended live evidence sequence

1	Boot and baseline	Record image versions, node health, management reachability, and synchronized time.
2	Core control plane	Prove OSPF, BFD, LDP, MPLS forwarding, MP-BGP, and route-reflector behavior.
3	Service paths	Validate enterprise, broadband, mobile, voice, and shared-Internet paths independently.
4	Security controls	Run positive and negative tests for routing policy, identity, trust, encryption, and management access.
5	Detection and recovery	Exercise selected failure scenarios, capture SIEM evidence, restore state, and verify cleanup.

PORTFOLIO CONCLUSION

The lab is a complete, internally consistent engineering package ready for controlled runtime execution. Its present public claim is intentionally narrower: validated architecture, security design, automation, and evidence planning.